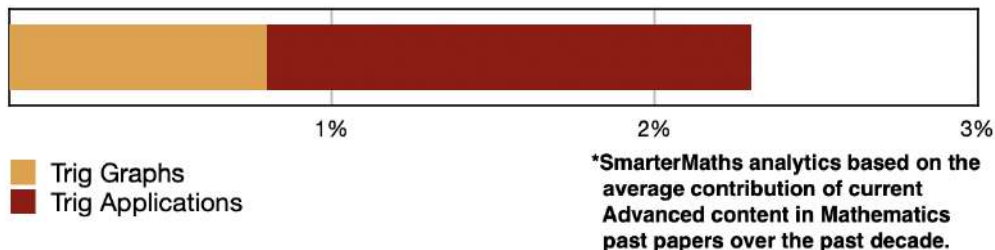


ADV: Trigonometry (Adv), T3 Trig Functions and Graphs (Adv)
Trig Graphs (Y12)
Trig Applications (Y12)

Teacher: Ljiljana Todic

Exam Equivalent Time: 138 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

T3 Trig Functions and Graphs



HISTORICAL CONTRIBUTION

- *T3 Trig Functions and Graphs* is a small contributor to past Mathematics exams, accounting for an average of 2.3% of past papers. We expect this to increase for reasons outlined below.
- This topic has been split into two sub-topics for analysis purposes: *1-Trig Graphs* (0.8%), and *2-Trig Applications* (1.5%).
- This analysis looks at the sub-topic *Trig Graphs*.

HSC ANALYSIS - What to expect and common pitfalls

- *Trig Graphs* have been examined in each of the last 5 years, receiving a multiple choice question on 4 occasions and a substantial 3 mark allocation in 2017.
- We expect *Trig Graphs* to be examined more often going forward due to the new syllabus content looking at transformations. This has been clearly flagged by the inclusion of 2 separate trig graph questions in the NESA sample HSC exam and our database, in response, has been significantly expanded in this area.
- We recommend close revision of *T3 EQ-Bank 3, 5* which are informed by the question style and difficulty level of NESA's sample questions. Also, careful attention should be given *2013 HSC 6 MC* which was surprisingly poorly answered.
- Note that more than half of students answered the 2016 multiple choice question on a tan function graph's *period* incorrectly. Deserves attention.

Questions

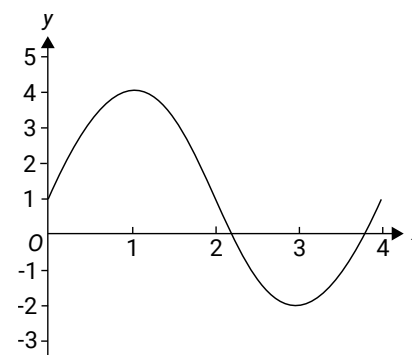
1. Trigonometry, 2ADV T3 2020 HSC 6 MC

Which interval gives the range of the function $y = 5 + 2 \cos 3x$?

- A. $[2, 8]$
- B. $[3, 7]$
- C. $[4, 6]$
- D. $[5, 9]$

2. Trigonometry, 2ADV T3 SM-Bank 8 MC

The diagram below shows one cycle of a circular function.



The amplitude and period of this function are respectively

- A. 3 and 2
- B. 3 and $\frac{\pi}{2}$
- C. 4 and $\frac{\pi}{4}$
- D. 3 and 4

3. Trigonometry, 2ADV T3 SM-Bank 1 MC

$$f(x) = 2\sin(3x) - 3$$

The period and range of this function are respectively

- (A) period = $\frac{2\pi}{3}$ and range = $[-5, -1]$
 - (B) period = $\frac{2\pi}{3}$ and range = $[-2, 2]$
 - (C) period = $\frac{\pi}{3}$ and range = $[-1, 5]$
 - (D) period = 3π and range = $[-1, 5]$
-

4. Trigonometry, 2ADV T3 SM-Bank 2 MC

Let $f(x) = 1 - 2\cos\left(\frac{\pi x}{2}\right)$.

The period and range of this function are respectively

- (A) 4 and $[-2, 2]$
 - (B) 4 and $[-1, 3]$
 - (C) 1 and $[-1, 3]$
 - (D) 4π and $[-2, 2]$
-

5. Trigonometry, 2ADV T3 SM-Bank 3 MC

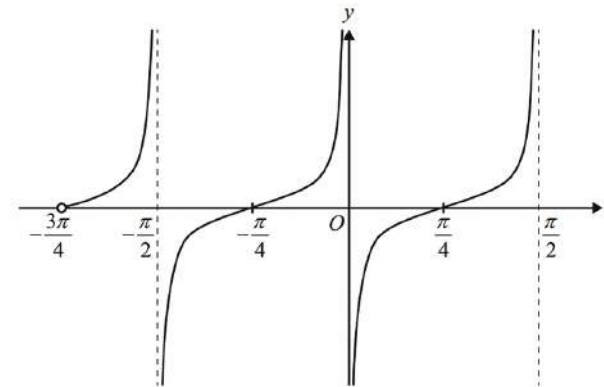
Let $f(x) = 5\sin(2x) - 1$.

The period and range of this function are respectively

- (A) π and $[-1, 4]$
 - (B) 2π and $[-1, 5]$
 - (C) π and $[-6, 4]$
 - (D) 2π and $[-6, 4]$
-

6. Trigonometry, 2ADV T3 SM-Bank 4 MC

A section of the graph of $f(x)$ is shown below.

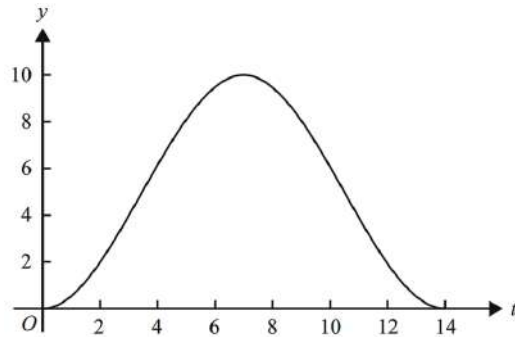


The equation of $f(x)$ could be

- (A) $f(x) = \tan(x)$
 - (B) $f(x) = \tan\left(x - \frac{\pi}{4}\right)$
 - (C) $f(x) = \tan\left(2\left(x - \frac{\pi}{4}\right)\right)$
 - (D) $f(x) = \tan\left(2\left(x - \frac{\pi}{2}\right)\right)$
-

7. Trigonometry, 2ADV T3 SM-Bank 5 MC

The UV index, y , for a summer day in Newcastle East is illustrated in the graph below, where t is the number of hours after 6 am.



The graph is most likely to be the graph of

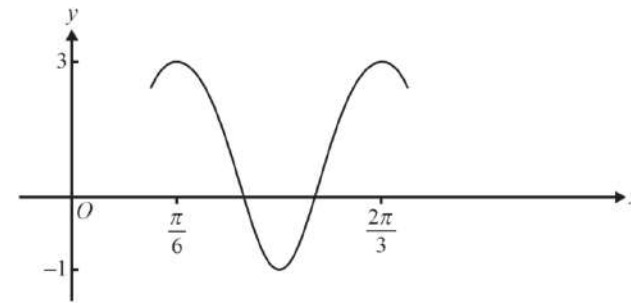
- (A) $y = 5 + 5 \cos\left(\frac{\pi t}{7}\right)$
- (B) $y = 5 - 5 \cos\left(\frac{\pi t}{7}\right)$
- (C) $y = 5 + 5 \cos\left(\frac{\pi t}{14}\right)$
- (D) $y = 5 - 5 \cos\left(\frac{\pi t}{14}\right)$

8. Trigonometry, 2ADV T3 SM-Bank 6 MC

The function with equation $f(x) = 4 \tan\left(\frac{x}{3}\right)$ has period

- (A) $\frac{2\pi}{3}$
- (B) 6π
- (C) 3
- (D) 3π

9. Trigonometry, 2ADV T3 SM-Bank 7 MC

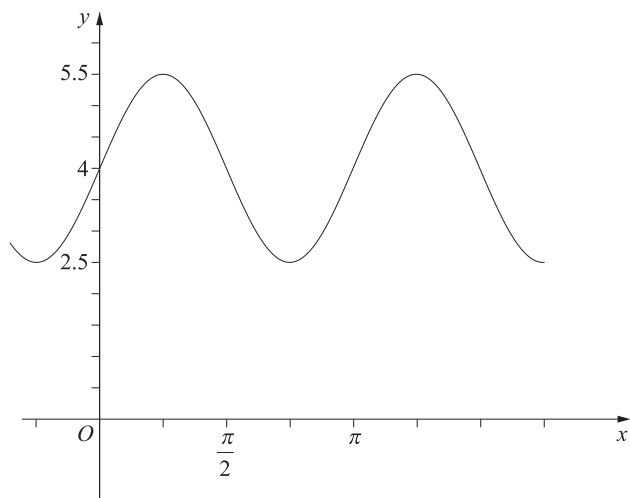


The graph shown could have equation

- (A) $y = 2 \cos\left(x + \frac{\pi}{6}\right) + 1$
- (B) $y = 2 \cos 4\left(x - \frac{\pi}{6}\right) + 1$
- (C) $y = 4 \sin 2\left(x - \frac{\pi}{12}\right) - 1$
- (D) $y = 3 \cos\left(2x + \frac{\pi}{6}\right) - 1$

10. Trigonometry, 2ADV T3 2019 HSC 7 MC

The diagram shows part of the graph of $y = a \sin(bx) + 4$.

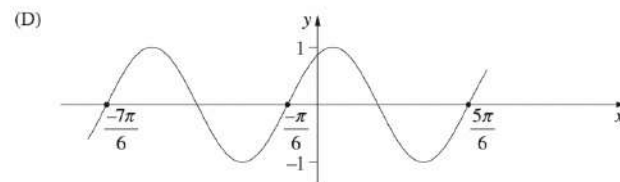
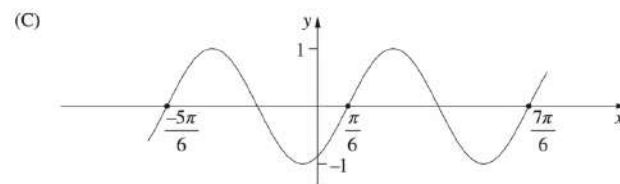
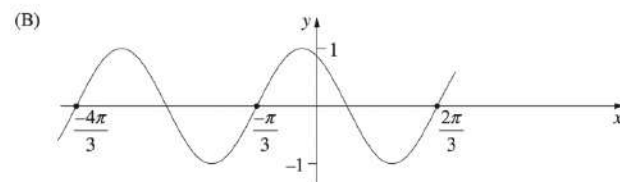
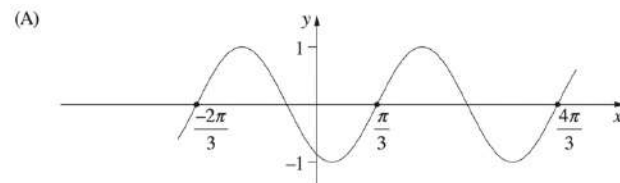


What are the values of a and b ?

- (A) $a = 3$ $b = \frac{1}{2}$
- (B) $a = 3$ $b = 2$
- (C) $a = 1.5$ $b = \frac{1}{2}$
- (D) $a = 1.5$ $b = 2$

11. Trigonometry, 2ADV T3 2013 HSC 6 MC

Which diagram shows the graph $y = \sin\left(2x + \frac{\pi}{3}\right)$?



12. Trigonometry, 2ADV T3 2016 HSC 6 MC

What is the period of the function $f(x) = \tan(3x)$?

- (A) $\frac{\pi}{3}$
- (B) $\frac{2\pi}{3}$
- (C) 3π
- (D) 6π

13. Trigonometry, 2ADV T3 2018 HSC 10 MC

A trigonometric function $f(x)$ satisfies the condition

$$\int_0^{\pi} f(x) dx \neq \int_{\pi}^{2\pi} f(x) dx.$$

Which function could be $f(x)$?

- (A) $f(x) = \sin(2x)$
- (B) $f(x) = \cos(2x)$
- (C) $f(x) = \sin\left(\frac{x}{2}\right)$
- (D) $f(x) = \cos\left(\frac{x}{2}\right)$

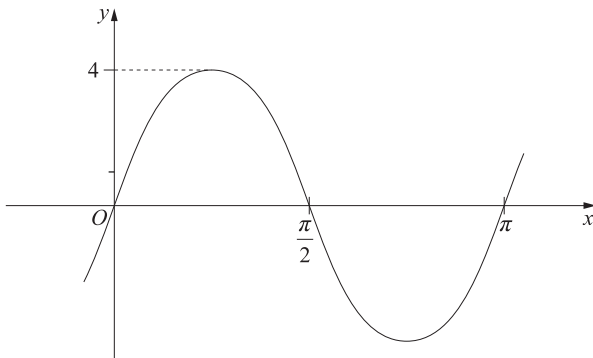
14. Trigonometry, 2ADV T3 EQ-Bank 5

The function $f(x) = \sin x$ is transformed into the function $g(x) = \frac{\sin(4x)}{3}$.

Describe in words how the amplitude and period have changed in this transformation. (2 marks)

15. Trigonometry, 2ADV T3 2010 HSC 8c

The graph shown is $y = A \sin bx$.



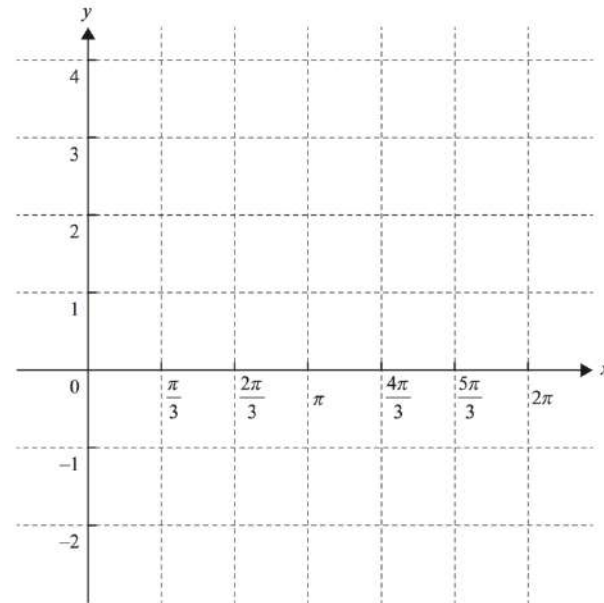
- i. Write down the value of A . (1 mark)
- ii. Find the value of b . (1 mark)
- iii. Copy or trace the graph into your writing booklet.

On the same set of axes, draw the graph $y = 3 \sin x + 1$ for $0 \leq x \leq \pi$. (2 marks)

16. Trigonometry, 2ADV T3 SM-Bank 9

Let $f(x) = 2 \cos(x) + 1$ for $0 \leq x \leq 2\pi$.

- i. Solve the equation $2 \cos(x) + 1 = 0$ for $0 \leq x \leq 2\pi$. (2 marks)
- ii. Sketch the graph of the function $f(x)$ on the axes below. Label the endpoints and local minimum point with their coordinates. (3 marks)



17. Trigonometry, 2ADV T3 2006 HSC 7b

A function $f(x)$ is defined by $f(x) = 1 + 2 \cos x$.

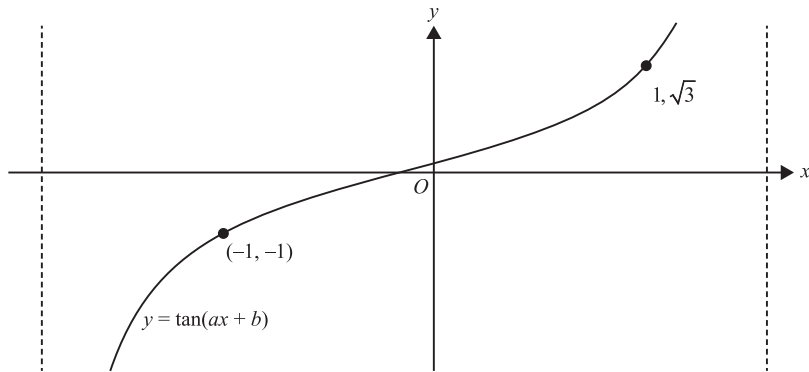
- i. Show that the graph of $y = f(x)$ cuts the x -axis at $x = \frac{2\pi}{3}$. (1 mark)
- ii. Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$ showing where the graph cuts each of the axes. (3 marks)
- iii. Find the area under the curve $y = f(x)$ between $x = -\frac{\pi}{2}$ and $x = \frac{2\pi}{3}$. (3 marks)

18. Trigonometry, 2ADV T3 2017 HSC 14a

Sketch the curve $y = 4 + 3 \sin 2x$ for $0 \leq x \leq 2\pi$. (3 marks)

19. Trigonometry, 2ADV T3 2010 MET1 3

Shown below is part of the graph of a period of the function of the form $y = \tan(ax + b)$.



Find the value of a and the value of b , where $a > 0$ and $0 < b < 1$. (3 marks)

20. Trigonometry, 2ADV T3 2021 HSC 20

For what values of x , in the interval $0 \leq x \leq \frac{\pi}{4}$, does the line $y = 1$ intersect the graph of $y = 2 \sin 4x$? (2 marks)

21. Trigonometry, 2ADV T3 SM-Bank 12

State the range and period of the function

$$h(x) = 4 + 3 \cos\left(\frac{\pi x}{2}\right). \quad (2 \text{ marks})$$

22. Trigonometry, 2ADV T3 SM-Bank 13

On any given day, the depth of water in a river is modelled by the function

$$h(t) = 14 + 8 \sin\left(\frac{\pi t}{12}\right), \quad 0 \leq t \leq 24$$

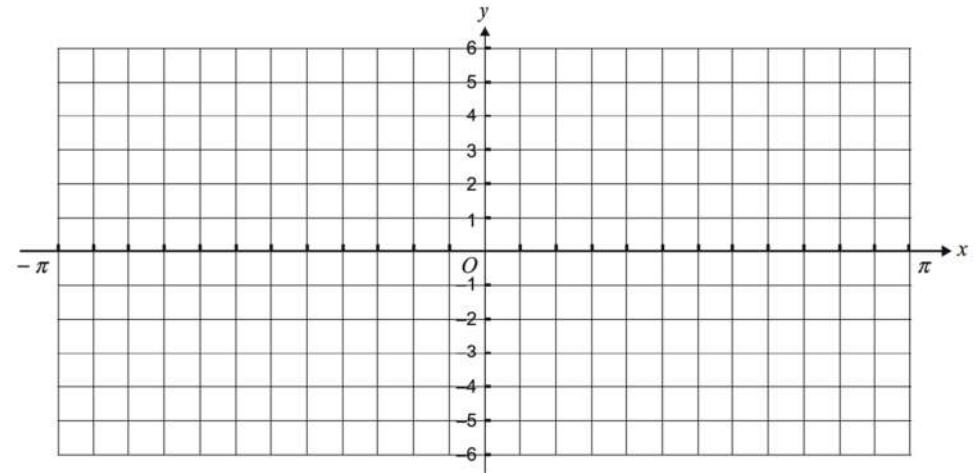
where h is the depth of water, in metres, and t is the time, in hours, after 6 am.

- Find the minimum depth of the water in the river. (1 mark)
- Find the values of t for which $h(t) = 10$. (2 marks)

23. Trigonometry, 2ADV T3 SM-Bank 14

For the function $f(x) = 5 \cos\left(2\left(x + \frac{\pi}{3}\right)\right)$, $-\pi \leq x \leq \pi$

- Write down the amplitude and period of the function (2 marks)
- Sketch the graph of the function $f(x)$ on the set of axes below. Label axes intercepts with their coordinates.
Label endpoints of the graph with their coordinates. (3 marks)



24. Trigonometry, 2ADV T3 SM-Bank 15

The graphs of $y = \cos(x)$ and $y = a \sin(x)$, where a is a real constant, have a point of intersection at $x = \frac{\pi}{3}$.

- Find the value of a . (2 marks)
- Find the x -coordinate of the other point of intersection of the two graphs, given $0 \leq x \leq 2\pi$ (1 mark)

25. Trigonometry, 2ADV T3 2018 HSC 15a

The length of daylight, $L(t)$, is defined as the number of hours from sunrise to sunset, and can be modelled by the equation

$$L(t) = 12 + 2 \cos\left(\frac{2\pi t}{366}\right),$$

where t is the number of days after 21 December 2015, for $0 \leq t \leq 366$.

- Find the length of daylight on 21 December 2015. (1 mark)
- What is the shortest length of daylight? (1 mark)
- What are the two values of t for which the length of daylight is 11? (2 marks)

26. Trigonometry, 2ADV T3 SM-Bank 10

The population of wombats in a particular location varies according to the rule

$n(t) = 1200 + 400 \cos\left(\frac{\pi t}{3}\right)$, where n is the number of wombats and t is the number of months after 1 March 2018.

- Find the period and amplitude of the function n . (2 marks)
- Find the maximum and minimum populations of wombats in this location. (2 marks)
- Find $n(10)$. (1 mark)
- Over the 12 months from 1 March 2018, find the fraction of time when the population of wombats in this location was less than $n(10)$. (2 marks)

27. Trigonometry, 2ADV T3 2013 HSC 13a

The population of a herd of wild horses is given by

$$P(t) = 400 + 50 \cos\left(\frac{\pi}{6}t\right)$$

where t is time in months.

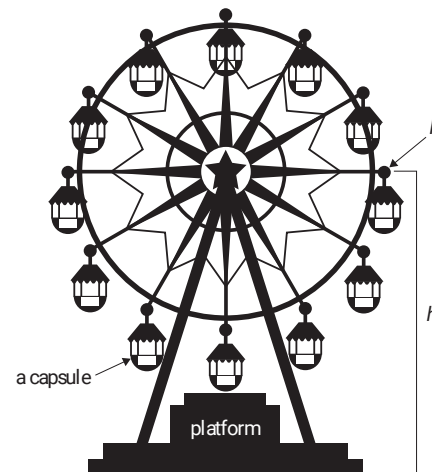
- Find all times during the first 12 months when the population equals 375 horses. (2 marks)
- Sketch the graph of $P(t)$ for $0 \leq t \leq 12$. (2 marks)

28. Trigonometry, 2ADV T3 SM-Bank 16

Sammy visits a giant Ferris wheel. Sammy enters a capsule on the Ferris wheel from a platform above the ground. The Ferris wheel is rotating anticlockwise. The capsule is attached to the Ferris wheel at point P . The height of P above the ground, h , is modelled by

$h(t) = 65 - 55 \cos\left(\frac{\pi t}{15}\right)$, where t is the time in minutes after Sammy enters the capsule and h is measured in metres.

Sammy exits the capsule after one complete rotation of the Ferris wheel.

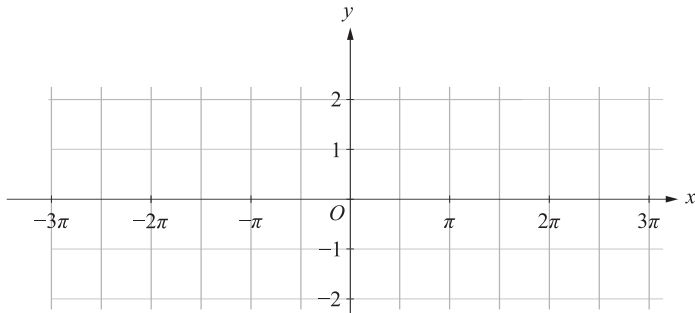


- State the minimum and maximum heights of P above the ground. (1 mark)
- For how much time is Sammy in the capsule? (1 mark)
- Find the rate of change of h with respect to t and, hence, state the value of t at which the rate of change of h is at its maximum. (2 marks)

29. Trigonometry, 2ADV T3 EQ-Bank 3

By drawing graphs on the number plane, show how many solutions exist for the equation

$$\cos x = \left| \frac{x - \pi}{4} \right| \text{ in the domain } (-\infty, \infty) \text{ (3 marks)}$$



30. Trigonometry, 2ADV T3 2011 SPEC1 8

Find the coordinates of the points of intersection of the graph of the relation

$$y = \operatorname{cosec}^2\left(\frac{\pi x}{6}\right) \text{ with the line } y = \frac{4}{3}, \text{ for } 0 < x < 12. \text{ (3 marks)}$$

31. Trigonometry, 2ADV T3 SM-Bank 8

$$f(x) = 2 \sin(2x) \text{ is defined in the domain } \left\{ x: \frac{\pi}{8} \leq x < \frac{\pi}{3} \right\}$$

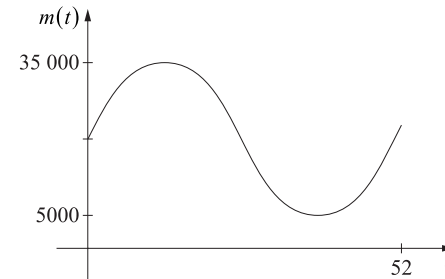
What is the range of the function $f(x)$? (2 marks)

32. Trigonometry, 2ADV T3 2020 HSC 31

The population of mice on an isolated island can be modelled by the function.

$$m(t) = a \sin\left(\frac{\pi}{26}t\right) + b,$$

where t is the time in weeks and $0 \leq t \leq 52$. The population of mice reaches a maximum of 35 000 when $t = 13$ and a minimum of 5000 when $t = 39$. The graph of $m(t)$ is shown.



a. What are the values of a and b ? (2 marks)

b. On the same island, the population of cats can be modelled by the function

$$c(t) = -80 \cos\left(\frac{\pi}{26}(t - 10)\right) + 120$$

Consider the graph of $m(t)$ and the graph of $c(t)$.

Find the values of t , $0 \leq t \leq 52$, for which both populations are increasing. (3 marks)

c. Find the rate of change of the mice population when the cat population reaches a maximum. (2 marks)

33. Trigonometry, 2ADV T3 2009 HSC 7b

Between 5 am and 5 pm on 3 March 2009, the height, h , of the tide in a harbour was given by

$$h = 1 + 0.7 \sin\left(\frac{\pi}{6}t\right) \text{ for } 0 \leq t \leq 12$$

where h is in metres and t is in hours, with $t = 0$ at 5 am.

i. What is the period of the function h ? (1 mark)

ii. What was the value of h at low tide, and at what time did low tide occur? (2 marks)

iii. A ship is able to enter the harbour only if the height of the tide is at least 1.35 m.

Find all times between 5 am and 5 pm on 3 March 2009 during which the ship was able to enter the harbour. (3 marks)

Worked Solutions

1. Trigonometry, 2ADV T3 2020 HSC 6 MC

$$\begin{aligned}-1 &\leq \cos 3x \leq 1 \\ -2 &\leq 2 \cos 3x \leq 2 \\ 3 &\leq 5 + 2 \cos 3x \leq 7 \\ \therefore \text{Range } [3, 7] \\ \Rightarrow B\end{aligned}$$

2. Trigonometry, 2ADV T3 SM-Bank 8 MC

$$\begin{aligned}\text{Graph centres around } y &= 1 \\ \text{Amplitude} &= 3 \\ \text{Period: } &= 4 \\ \Rightarrow D\end{aligned}$$

3. Trigonometry, 2ADV T3 SM-Bank 1 MC

$$\begin{aligned}\text{Range: } [-3 - 2, -3 + 2] \\ &= [-5, -1] \\ \text{Period} &= \frac{2\pi}{n} = \frac{2\pi}{3} \\ \Rightarrow A\end{aligned}$$

4. Trigonometry, 2ADV T3 SM-Bank 2 MC

$$\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{\frac{\pi}{2}} = 4$$

$$\begin{aligned}\text{Amplitude} &= 2 \\ \text{Graph centre line (median): } y &= 1. \\ \therefore \text{Range} &= [1 - 2, 1 + 2] \\ &= [-1, 3] \\ \Rightarrow B\end{aligned}$$

5. Trigonometry, 2ADV T3 SM-Bank 3 MC

$$\begin{aligned}\text{Period} &= \frac{2\pi}{2} = \pi \\ \text{Range} &= [-1 - 5, -1 + 5] \\ &= [-6, 4] \\ \Rightarrow C\end{aligned}$$

6. Trigonometry, 2ADV T3 SM-Bank 4 MC

$$\begin{aligned}\text{Period} &= \frac{\pi}{2} \\ \Rightarrow \text{must be } C \text{ or } D \\ \text{Shift } y = \tan(x) \text{ right } &\frac{\pi}{4}. \\ \Rightarrow C\end{aligned}$$

7. Trigonometry, 2ADV T3 SM-Bank 5 MC

$$\begin{aligned}\text{Centre line (median): } y &= 5 \\ \text{Amplitude} &= 5 \\ \text{Period: } 14 &= \frac{2\pi}{n} \\ n &= \frac{\pi}{7} \\ \therefore \text{Graph: } y &= 5 - 5 \cos\left(\frac{\pi t}{7}\right) \\ \Rightarrow B\end{aligned}$$

8. Trigonometry, 2ADV T3 SM-Bank 6 MC

$$\begin{aligned}\text{Period} &= \frac{\pi}{n} \\ &= \frac{\pi}{\frac{1}{3}} \\ &= 3\pi \\ \Rightarrow D\end{aligned}$$

9. Trigonometry, 2ADV T3 SM-Bank 7 MC

Amplitude = 2 (range from -1 to 3)

Graph centre line (median): $y = 1$

$\therefore$ Eliminate C and D .

$$\text{Period} = \frac{2\pi}{3} - \frac{\pi}{6} = \frac{\pi}{2} \text{ (from graph)}$$

Consider option B ,

$$\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{4} = \frac{\pi}{2}$$

$\Rightarrow B$

10. Trigonometry, 2ADV T3 2019 HSC 7 MC

$$a = \frac{1}{2}(5.5 - 2.5) = 1.5$$

Since graph passes through $\left(\frac{\pi}{4}, 5.5\right)$:

$$5.5 = 1.5 \sin\left(b \times \frac{\pi}{4}\right) + 4$$

$$\sin\left(b \times \frac{\pi}{4}\right) = 1$$

$$b \times \frac{\pi}{4} = \frac{\pi}{2}$$

$$\therefore b = 2$$

$\Rightarrow D$

11. Trigonometry, 2ADV T3 2013 HSC 6 MC

$$\text{At } x = 0, y = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$\Rightarrow$ It cannot be A or C

Find x when $y = 0$,

$$\sin\left(2x + \frac{\pi}{3}\right) = 0$$

$$\therefore 2x + \frac{\pi}{3} = 0 \quad (\sin 0 = 0)$$

$$2x = -\frac{\pi}{3}$$

$$x = -\frac{\pi}{6}$$

$\Rightarrow D$

12. Trigonometry, 2ADV T3 2016 HSC 6 MC

$$\text{Period} = \frac{\pi}{n}$$

$$= \frac{\pi}{3}$$

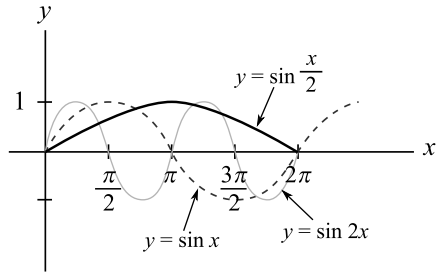
$\Rightarrow A$

♦♦ Mean mark 34%

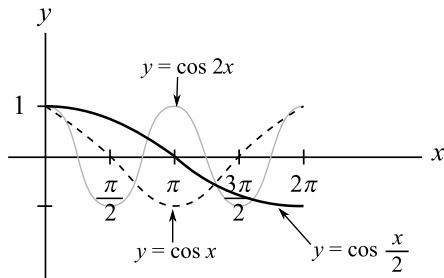
♦ Mean mark 42%

13. Trigonometry, 2ADV T3 2018 HSC 10 MC

Consider options A and C



Consider options B and D



When $y = \cos \frac{x}{2}$,

$$\int_0^{\pi} f(x) dx \neq \int_{\pi}^{2\pi} f(x) dx$$

$$\Rightarrow D$$

14. Trigonometry, 2ADV T3 EQ-Bank 5

$$g(x) = \frac{1}{3} \sin(4x)$$

$\Rightarrow$ The new amplitude is one third of the original amplitude.

$$\text{Period} = \frac{2\pi}{n} \Rightarrow n = \frac{1}{4}$$

$\Rightarrow$ The new period is one quarter of the original period.

15. Trigonometry, 2ADV T3 2010 HSC 8c

i. $A = 4$

ii. Since the graph passes through $\left(\frac{\pi}{4}, 4\right)$

Substituting into $y = 4 \sin bx$

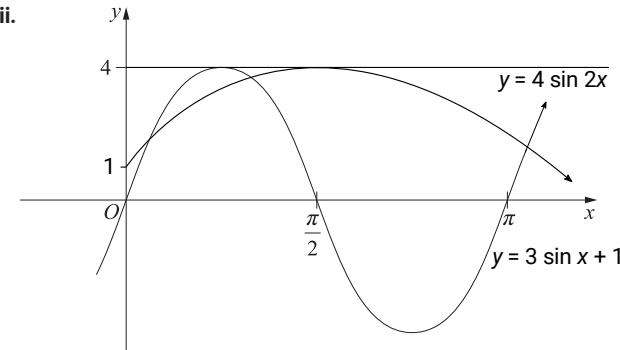
$$4 \sin\left(b \times \frac{\pi}{4}\right) = 4$$

$$\sin\left(b \times \frac{\pi}{4}\right) = 1$$

$$b \times \frac{\pi}{4} = \frac{\pi}{2}$$

$$\therefore b = 2$$

iii.



MARKER'S COMMENT: Graphs are consistently drawn too small by many students. Aim to make your diagrams 1/3 to 1/2 of a page.

16. Trigonometry, 2ADV T3 SM-Bank 9

i. $2 \cos(x) + 1 = 0$

$$\cos(x) = -\frac{1}{2}$$

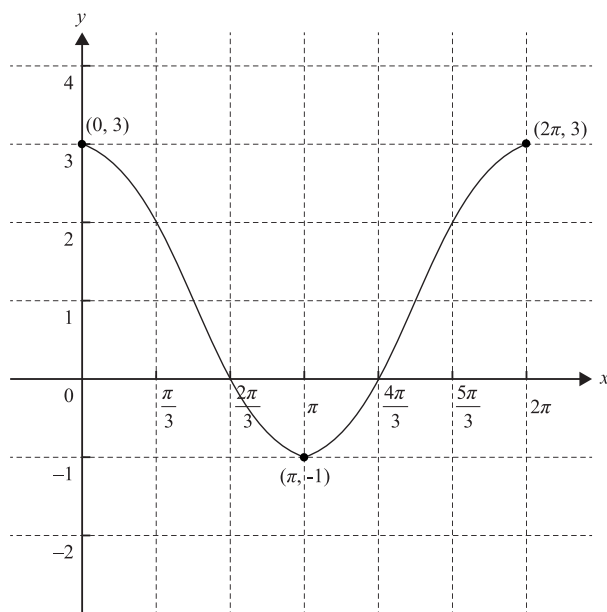
$$\Rightarrow \cos \frac{\pi}{3} = \frac{1}{2} \text{ and } \cos \text{ is negative}$$

in 2nd/3rd quadrant

$$\therefore x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$$

$$= \frac{2\pi}{3}, \frac{4\pi}{3}$$

ii.



17. Trigonometry, 2ADV T3 2006 HSC 7b

i. $f(x) = 1 + 2 \cos x$

$f(x)$ cuts the x -axis when $f(x) = 0$

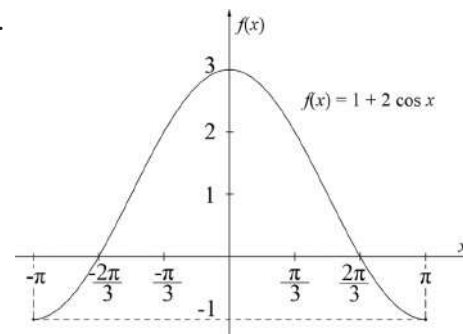
$$1 + 2 \cos x = 0$$

$$2 \cos x = -1$$

$$\cos x = -\frac{1}{2}$$

$$\therefore x = \frac{2\pi}{3} \dots \text{as required}$$

ii.



iii. Area = $\int_{-\frac{\pi}{2}}^{\frac{2\pi}{3}} 1 + 2 \cos x \, dx$

$$= [x + 2 \sin x]_{-\frac{\pi}{2}}^{\frac{2\pi}{3}}$$

$$= \left[\left(\frac{2\pi}{3} + 2 \sin \frac{2\pi}{3} \right) - \left(-\frac{\pi}{2} + 2 \sin -\frac{\pi}{2} \right) \right]$$

$$= \left(\frac{2\pi}{3} + 2 \times \frac{\sqrt{3}}{2} \right) - \left(-\frac{\pi}{2} + 2(-1) \right)$$

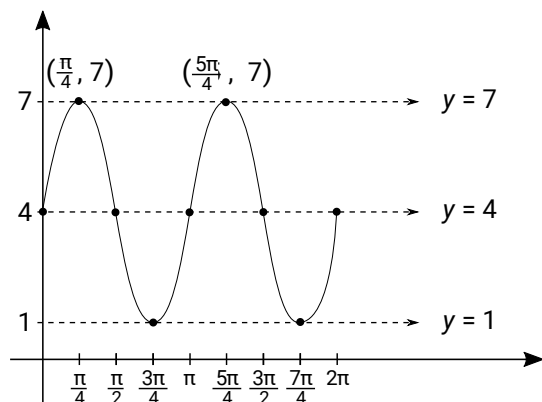
$$= \frac{2\pi}{3} + \sqrt{3} + \frac{\pi}{2} + 2$$

$$= \left(\frac{7\pi}{6} + \sqrt{3} + 2 \right) \text{ u}^2$$

18. Trigonometry, 2ADV T3 2017 HSC 14a

$$y = 4 + 3 \sin 2x$$

$\Rightarrow$ Amplitude of 3 about $y = 4$



19. Trigonometry, 2ADV T3 2010 MET1 3

$$y = \tan(ax + b)$$

Substitute $(1, \sqrt{3})$, $(-1, -1)$ into equation:

$$\tan(a + b) = \sqrt{3}$$

$$\tan(b - a) = -1$$

$$a + b = \frac{\pi}{3} \dots (1)$$

$$b - a = -\frac{\pi}{4} \dots (2)$$

Add (1) + (2):

$$2b = \frac{\pi}{3} - \frac{\pi}{4}$$

$$b = \frac{\pi}{24}$$

Substitute into (1):

$$a + \frac{\pi}{24} = \frac{\pi}{3}$$

$$a = \frac{7\pi}{24}$$

20. Trigonometry, 2ADV T3 2021 HSC 20

Find x such that:

$$2 \sin 4x = 1$$

$$\sin 4x = \frac{1}{2}$$

$$4x = \sin^{-1} \frac{1}{2}$$

$$4x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \dots$$

$$\therefore x = \frac{\pi}{24}, \frac{5\pi}{24} \quad \left(0 \leq x \leq \frac{\pi}{4}\right)$$

21. Trigonometry, 2ADV T3 SM-Bank 12

$$-1 \leq \cos\left(\frac{\pi x}{2}\right) \leq 1$$

$$-3 \leq 3 \cos\left(\frac{\pi x}{2}\right) \leq 3$$

$$1 \leq 4 + 3 \cos\left(\frac{\pi x}{2}\right) \leq 7$$

$$\therefore \text{Range: } 1 \leq y \leq 7$$

$$\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{\frac{\pi}{2}} = 4$$

22. Trigonometry, 2ADV T3 SM-Bank 13

i. $h_{\min}$ occurs when $\sin\left(\frac{\pi t}{12}\right) = -1$

$$\therefore h_{\min} = 14 - 8 \\ = 6 \text{ m}$$

MARKER'S COMMENT: Students who used calculus to find the minimum were less successful.

ii. $14 + 8 \sin\left(\frac{\pi}{12}t\right) = 10$

$$\sin\left(\frac{\pi}{12}t\right) = -\frac{1}{2}$$

Solve in general:

$$\frac{\pi}{12}t = \frac{7\pi}{6} + 2\pi n \quad \text{or} \quad \frac{\pi}{12}t = \frac{11\pi}{6} + 2\pi n, \\ t = 14 + 24n \quad \quad \quad t = 22 + 24n$$

Substitute integer values for n ,

$$\therefore t = 14 \text{ or } 22, \quad (0 \leq t \leq 24)$$

23. Trigonometry, 2ADV T3 SM-Bank 14

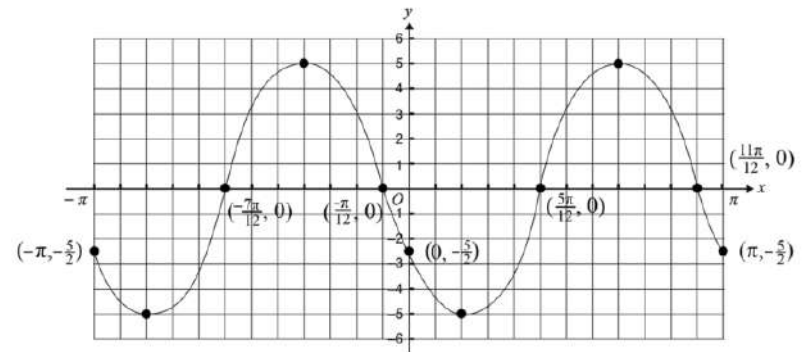
a. Amplitude = 5

$$\text{Period} = \frac{2\pi}{2} = \pi$$

b. Shift $y = 5 \cos(2x)$ left $\frac{\pi}{3}$ units.

$$\text{Period} = \pi$$

$$\text{Endpoints are } \left(-\pi, -\frac{5}{2}\right) \text{ and } \left(\pi, -\frac{5}{2}\right)$$



24. Trigonometry, 2ADV T3 SM-Bank 15

i. Intersection occurs when $x = \frac{\pi}{3}$,

$$a \sin\left(\frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right)$$

$$\tan\left(\frac{\pi}{3}\right) = \frac{1}{a}$$

$$\sqrt{3} = \frac{1}{a}$$

$$\therefore a = \frac{1}{\sqrt{3}}$$

ii. $\tan(x) = \sqrt{3}$

$$x = \frac{\pi}{3}, \frac{4\pi}{3}, 2\pi + \frac{\pi}{3}, \dots$$

$$\therefore x = \frac{4\pi}{3} \quad (0 \leq x \leq 2\pi)$$

25. Trigonometry, 2ADV T3 2018 HSC 15a

i. $L(t) = 12 + 2 \cos\left(\frac{2\pi t}{366}\right)$

On 21 Dec 2015 $\Rightarrow t = 0$

$$\therefore L(0) = 12 + 2 \cos 0$$

$$= 14 \text{ hours}$$

ii. Shortest length of daylight occurs when

$$\cos\left(\frac{2\pi t}{366}\right) = -1$$

$$\therefore \text{Shortest length} = 12 + 2(-1)$$

$$= 10 \text{ hours}$$

iii. Find t such that $L(t) = 11$:

$$11 = 12 + 2 \cos\left(\frac{2\pi t}{366}\right)$$

$$\cos\left(\frac{2\pi t}{366}\right) = -\frac{1}{2}$$

$$\frac{2\pi t}{366} = \frac{2\pi}{3} \quad \text{or} \quad \frac{2\pi t}{366} = \frac{4\pi}{3}$$

$$t = \frac{366}{3} \quad t = \frac{366 \times 2}{3}$$

$$= 122 \quad = 244$$

$$\therefore t = 122 \text{ or } 244$$

♦ Mean mark 43%.

26. Trigonometry, 2ADV T3 SM-Bank 10

i. $\text{Period} = \frac{2\pi}{n} = \frac{2\pi}{\frac{\pi}{3}} = 6 \text{ months}$

$\text{Amplitude} = 400$

ii. $\text{Max: } 1200 + 400 = 1600 \text{ wombats}$

$\text{Min: } 1200 - 400 = 800 \text{ wombats}$

iii. $n(10) = 1200 + 400 \cos\left(\frac{10\pi}{3}\right)$

$$= 1200 + 400 \cos\left(\frac{2\pi}{3}\right)$$

$$= 1200 - 400 \times \frac{1}{2}$$

$$= 1000 \text{ wombats}$$

iv. Find t when $n(t) = 1000$

$$1000 = 1200 + 400 \cos\left(\frac{\pi t}{3}\right)$$

$$\cos\left(\frac{\pi t}{3}\right) = -\frac{1}{2}$$

$$\frac{\pi t}{3} = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}, \dots$$

$$t = 2, 4, 8, 10$$

Since $n(0) = 1600$,

$\Rightarrow n(t)$ drops below 1000 between $t = 2$ and $t = 4$,
and between $t = 8$ and $t = 10$.

$$\begin{aligned} \therefore \text{Fraction} &= \frac{2+2}{12} \\ &= \frac{1}{3} \text{ year} \end{aligned}$$

27. Trigonometry, 2ADV T3 2013 HSC 13a

i. $P(t) = 400 + 50 \cos\left(\frac{\pi}{6}t\right)$

Need to find t when $P(t) = 375$

$$375 = 400 + 50 \cos\left(\frac{\pi}{6}t\right)$$

$$50 \cos\left(\frac{\pi}{6}t\right) = -25$$

$$\cos\left(\frac{\pi}{6}t\right) = -\frac{1}{2}$$

Since $\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$, and $\cos$ is

negative in 2nd / 3rd quadrants:

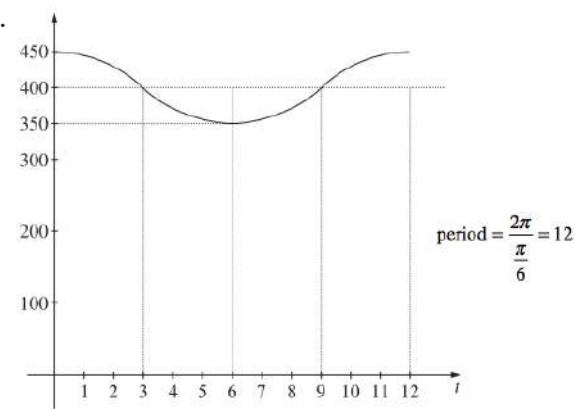
$$\Rightarrow \frac{\pi}{6}t = \left(\pi - \frac{\pi}{3}\right), \left(\pi + \frac{\pi}{3}\right), \left(3\pi - \frac{\pi}{3}\right)$$

$$= \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \dots$$

$$\therefore t = 4, 8, 16, \dots$$

$\therefore$ In the 1st 12 months, $P(t) = 375$ when
 $t = 4$ months and 8 months.

ii.



♦ Mean mark 39%

28. Trigonometry, 2ADV T3 SM-Bank 16

i. $h_{\min} = 65 - 55$ $h_{\max} = 65 + 55$
 $= 10 \text{ m}$ $= 120 \text{ m}$

ii. $\text{Period} = \frac{2\pi}{\frac{\pi}{15}} = 30 \text{ min}$

iii. $h'(t) = 65 - 55 \cos\left(\frac{\pi t}{15}\right)$

$$h'(t) = \frac{\pi}{15} \times 55 \sin\left(\frac{\pi}{15}t\right)$$

$$= \frac{11\pi}{3} \sin\left(\frac{\pi}{15}t\right)$$

Since $\sin\left(\frac{\pi}{15}t\right)_{\max} = \sin\left(\frac{\pi}{2}\right)$,

$\therefore h'(t)_{\max}$ occurs when

$$\frac{\pi t}{15} = \frac{\pi}{2}$$

$$\therefore t = \frac{\pi}{2} \times \frac{15}{\pi}$$

$$= \frac{15}{2} \text{ minutes } (0 \leq t \leq 30)$$

29. Trigonometry, 2ADV T3 EQ-Bank 3

Sketch:

$$y = \cos x$$

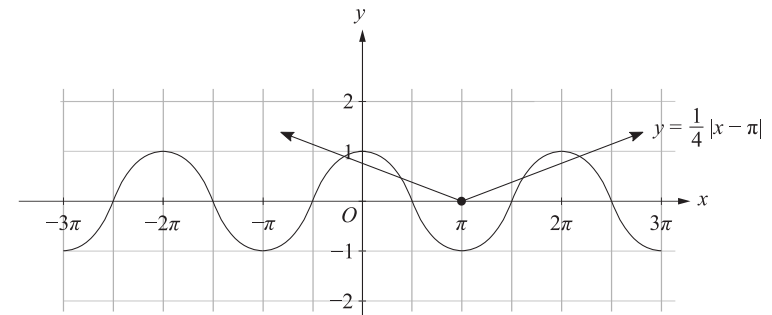
$$y = \left| \frac{x - \pi}{4} \right|$$

Translate π units to the right:

$$y = |x| \Rightarrow y = |x - \pi|$$

Multiply by $\frac{1}{4}$:

$$y = |x - \pi| \Rightarrow y = \frac{1}{4}|x - \pi| = \left| \frac{x - \pi}{4} \right|$$



$\therefore$ There are 4 solutions.

30. Trigonometry, 2ADV T3 2011 SPEC1 8

Intersection occurs when:

$$\operatorname{cosec}^2\left(\frac{\pi x}{6}\right) = \frac{4}{3}$$

$$\operatorname{cosec}\left(\frac{\pi x}{6}\right) = \pm \frac{2}{\sqrt{3}}$$

$$\sin\left(\frac{\pi x}{6}\right) = \pm \frac{\sqrt{3}}{2}$$

$$\text{Given: } 0 < x < 12 \Rightarrow 0 < \frac{\pi x}{6} < 2\pi$$

$$\frac{\pi x}{6} = \frac{\pi}{3}, \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$= \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

$$x = 2, 4, 8, 10$$

$$\Rightarrow y = \frac{4}{3} \text{ for each}$$

$$\therefore \text{Intersection at: } \left(2, \frac{4}{3}\right), \left(4, \frac{4}{3}\right), \left(8, \frac{4}{3}\right), \left(10, \frac{4}{3}\right)$$

31. Trigonometry, 2ADV T3 SM-Bank 8

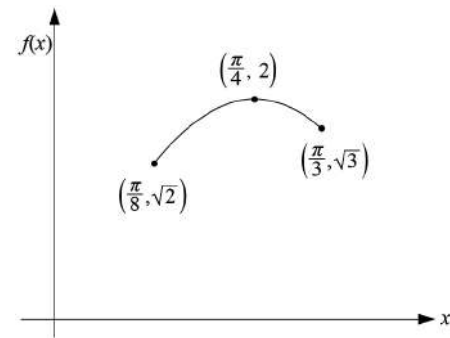
$$\sin(2x)_{\max} \text{ occurs when } x = \frac{\pi}{4} \text{ (within domain)}$$

$$\Rightarrow f(x)_{\max} = 2 \sin\left(\frac{\pi}{2}\right) = 2$$

Checking endpoints:

$$\text{When } x = \frac{\pi}{8} \Rightarrow y = 2 \sin\left(\frac{\pi}{4}\right) = \sqrt{2}$$

$$\text{When } x = \frac{\pi}{3} \Rightarrow y = 2 \sin\left(\frac{2\pi}{3}\right) = \sqrt{3}$$



$$\therefore \text{Range} = [\sqrt{2}, 2],$$

♦ Mean mark 45%.

32. Trigonometry, 2ADV T3 2020 HSC 31

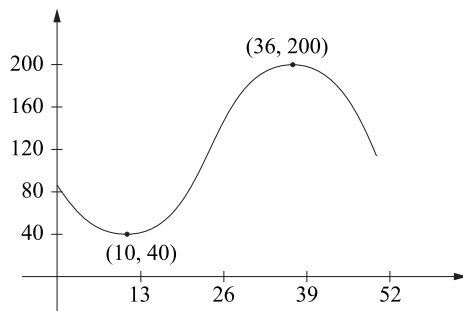
$$\begin{aligned} \text{a. } b &= \frac{35\,000 + 5000}{2} \\ &= 20\,000 \end{aligned}$$

$$\begin{aligned} a &= \text{amplitude of sin graph} \\ &= 35\,000 - 20\,000 \\ &= 15\,000 \end{aligned}$$

b. By inspection of the $m(t)$ graph

$$m'(t) > 0 \text{ when } 0 \leq t < 13 \text{ and } 39 < t \leq 52$$

Sketch $c(t)$:



Minimum ($\cos 0$) when $t = 10$

Maximum ($\cos \pi$) when $t = 36$

$$\therefore c'(t) > 0 \text{ when } 10 < t < 36$$

$\therefore$ Both populations are increasing when $10 < t < 36$

♦♦ Mean mark part (b) 30%.

c. $c(t)$ maximum when $t = 36$

$$m(t) = 15\,000 \sin\left(\frac{\pi}{26}t\right) + 20\,000$$

$$m'(t) = \frac{15\,000\pi}{26} \cos\left(\frac{\pi}{26}t\right)$$

$$\begin{aligned} m'(36) &= \frac{15\,000\pi}{26} \cdot \cos\left(\frac{36\pi}{26}\right) \\ &= -642.7 \end{aligned}$$

$\therefore$ Mice population is decreasing at 643 mice per week.

♦♦♦ Mean mark part (c) 27%.

33. Trigonometry, 2ADV T3 2009 HSC 7b

$$\text{i. } h = 1 + 0.7 \sin\left(\frac{\pi}{6}t\right) \text{ for } 0 \leq t \leq 12$$

$$\begin{aligned} T &= \frac{2\pi}{n} \text{ where } n = \frac{\pi}{6} \\ &= 2\pi \times \frac{6}{\pi} \\ &= 12 \text{ hours} \end{aligned}$$

$\therefore$ The period of h is 12 hours.

ii. Find h at low tide

$\Rightarrow h$ will be a minimum when

$$\sin\left(\frac{\pi}{6}t\right) = -1$$

$$\begin{aligned} \therefore h_{\min} &= 1 + 0.7(-1) \\ &= 0.3 \text{ metres} \end{aligned}$$

$$\text{Since } \sin x = -1 \text{ when } x = \frac{3\pi}{2}$$

$$\begin{aligned} \frac{\pi}{6}t &= \frac{3\pi}{2} \\ t &= \frac{3\pi}{2} \times \frac{6}{\pi} \\ &= 9 \text{ hours} \end{aligned}$$

$\therefore$ Low tide occurs at 2pm (5 am + 9 hours)

iii. Find t when $h \geq 1.35$

$$1 + 0.7 \sin\left(\frac{\pi}{6}t\right) \geq 1.35$$

$$0.7 \sin\left(\frac{\pi}{6}t\right) \geq 0.35$$

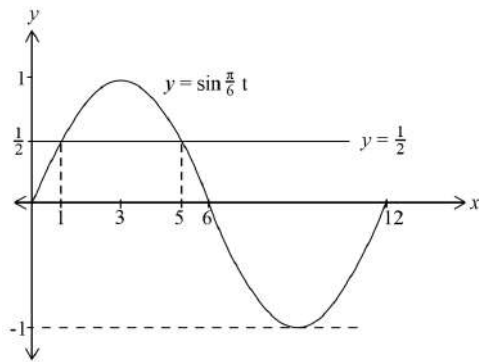
$$\sin\left(\frac{\pi}{6}t\right) \geq \frac{1}{2}$$

$$\sin\left(\frac{\pi}{6}t\right) = \frac{1}{2} \text{ when}$$

$$\frac{\pi}{6}t = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \text{ etc ...}$$

IMPORTANT: Using $\sin x = -1$ for a minimum here is very effective and time efficient. This property of trig functions is **often very useful** in harder questions.

$$t = 1, 5 \quad (0 \leq t \leq 12)$$



From the graph,

$$\sin\left(\frac{\pi}{6}t\right) \geq \frac{1}{2} \quad \text{when } 1 \leq t \leq 5$$

$\therefore$ Ship can enter the harbour between 6 am and 10 am.